INSTITUTO UNIVERSITÁRIO DE LISBOA

Master's in Data Science

Course Unit: Programming Fundamentals

Lecturers: João Moreira and Tiago Prata

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Real Estate Lisboa

Group no. 2:
André Marques, no. 136447, (30%)
Fernando Rocha, no. 134110, (37%)
Frederico Ferreira, no. 135644, (33%)

Group no. 2: André Marques, no. 136447; Fernando Rocha, no. 134110; Frederico Ferreira, no. 135644. ISCTE Business School

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Real Estate Lisboa

Domain

This project focuses on the Portuguese residential real estate market, with an emphasis on the district of Lisbon. This choice is justified by Lisbon's economic relevance, as it has the highest population density in the country and the highest property transaction values.

The central objective is to understand the main factors that influence the sale price of homes intended for permanent residence, namely apartments and *duplexes*. We excluded non-comparable property types such as detached houses, land, shops or industrial spaces, ensuring greater consistency in the predictive modelling. The variables used reflect the typical structure of real estate listings: price, area, number of rooms, among other attributes that make it possible to identify relevant patterns for estimating the sale price.

Data Source

The database used in this work is the 2024 “Portuguese Real Estate Dataset”, publicly available on *Kaggle* and compiled by Gonçalo Baião. The set contains more than 120 thousand listings of properties for sale in Portugal, which makes it suitable both for exploratory analyses and for building predictive models.

The *dataset* includes the information usually found in real estate listings: the property price, the total area, the number of rooms, the number of bathrooms, the property type and its location (district and municipality), as can be seen in Annex 1. These elements make it possible to characterise each property in some detail and, at the same time, to adapt the dataset to the objective of the project. In our case, we applied a filter to keep only the properties in the district of Lisbon and to limit the analysis to the *Apartment* and *Duplex* types, ensuring that we work only with properties intended for urban housing.

The file is provided in CSV format, which makes it easy to import and process the data using libraries such as “pandas”.

Data Acquisition

The data were obtained directly from *Kaggle* through the *kagglehub* library, ensuring the most recent version available on the platform. The *dataset* was loaded into a *DataFrame* with pandas, manually adjusting the types of some columns with inconsistencies to ensure a correct import.

Data Processing

We selected a subset of attributes based on their theoretical relationship with price and on the prevalence of non-null values: "Price", "District", "City", “Town”,

"TotalArea", "Type", "HasParking", "ConstructionYear", "HasElevator", "TotalRooms" and "NumberOfBathrooms".

We filtered the data to keep only records from the district of Lisbon. This methodological decision was based on the fact that the Portuguese real estate market is not homogeneous: the price of an apartment in Lisbon and in an inland district follow completely different logics. Although the Lisbon market itself is not internally homogeneous, delimiting the analysis reduces exogenous variability and allows the model to capture the patterns specific to this market, resulting in more accurate predictions.

We corrected the HasParking and HasElevator columns, converting the text "True" to a boolean type, since the original dataset stored these values as strings and scikit-learn models require numerical or logical inputs.

We limited the analysis to the "Apartment" and "Duplex" types, excluding categories with different valuation logics (shops, land, detached houses) that would introduce noise into the model.

We created the "Age" column by subtracting ConstructionYear from 2024, establishing a more direct and interpretable relationship with price: an increase in the age of the property is associated with a decrease in price.

Next, we excluded rows where Price was null, since in a supervised learning model it is impossible to train without the actual price value. After analysing the percentiles of the total area, we set the upper limit at 10 000 m^2 to maintain the integrity of the dataset. We therefore applied filters to keep records where TotalArea was between 20 and 10 000 m^2 and NumberOfBathrooms was greater than or equal to zero, removing clearly incorrect data. An example of the cleaned dataset can be found in Annex 2.

Building the Predictive Model

The objective of the predictive model is to estimate the sale value of residential properties in the district of Lisbon, which makes it a supervised regression problem. We defined Price as the target variable and categorised the attributes as numerical (TotalArea, Age, TotalRooms, NumberOfBathrooms), categorical (City, Town, Type) and boolean (HasParking, HasElevator).

The modelling was implemented through a scikit-learn pipeline that encapsulates all the pre-processing steps, ensuring that the training and test data are transformed identically. For numerical attributes, we used SimpleImputer with the median; for categorical attributes, SimpleImputer with the mode followed by OneHotEncoder; for boolean attributes, we kept the original values.

As the estimator, we selected the RandomForestRegressor for its predictive power, robustness to outliers and ability to capture complex non-linear relationships. The dataset was split into training (80%) and test (20%) sets. The model was trained on the training set and its performance was evaluated on the test set to assess its generalisation ability.

For evaluation, we used R^2 (Coefficient of Determination), MAE (Mean Absolute Error), RMSE (Root Mean Squared Error) and MAPE (Mean Absolute Percentage Error). The model achieved an R^2 of 0.89, an MAE of 70 818.69 €, an RMSE of 153 036.23 € and a MAPE of 15.13%, as can be seen in Annex 3.

The results show a consistent model, with a good level of explanation of price variability and error values within what is expected for the Lisbon market.

Data Visualisation

For the exploratory analysis of the dataset, five visualisations were generated using the matplotlib library, with the aim of understanding the structure of the data and the relationships between the variables.

The bar chart (Annex 4) compares the average price by property type. This type of chart is ideal for comparisons between discrete categories, as it allows immediate visual perception through the alignment and length of the bars. The analysis reveals that duplexes have a higher average value than apartments.

The scatter plot (Annex 5) examines the relationship between total area and price. This type of chart is the standard tool for visualising correlations between two continuous variables, making it possible to identify patterns and relationships. The logarithmic scale on both axes (log-log) normalises the skewed distributions of both variables, revealing a linear relationship between the logarithm of the area and the logarithm of the price.

The histogram (Annex 6) analyses the distribution of the target variable on its own. This type of visualisation is essential for understanding the shape of the distribution of numerical variables. The logarithmic scale on the X axis compensates for the strong skewness of the price data, making it possible to observe the full range of values without outliers distorting the visualisation.

The boxplot (Annex 7) compares the distribution of prices across cities. This type of chart is superior to simple bar charts because it presents a complete statistical summary (median, quartiles, outliers) for each category, allowing comparisons of distributions and not just of means. The logarithmic scale on the Y axis makes the distributions comparable across cities with very different price levels.

Finally, the correlation matrix (Annex 8) analyses the linear relationships between numerical variables. This type of visualisation is fundamental for quickly identifying which attributes are most strongly associated with the target variable. The analysis reveals that the number of bathrooms has the strongest correlation with price, followed by the total area and the total number of rooms. The age of the property shows a moderate negative correlation, confirming that older properties tend to have lower prices.

Conclusion

This project made it possible to create a predictive model to estimate the prices of residential properties in the district of Lisbon, obtaining consistent

performance and reaching a level of performance that combines a good ability to explain price variation with a relatively low average error, in a particularly varied market that is difficult to model. One of the aspects we found most interesting was the exploration of the different data visualisations, which helped us better understand the patterns and specific characteristics of the Lisbon urban real estate market.

The greatest difficulty arose in handling the outliers and inconsistent values present in the dataset, particularly in defining reasonable limits for the total area. This process required several iterations and a careful analysis of the distributions, in order to ensure that the final dataset was consistent and that the model produced reliable results.

Annexes

Annex 1 - Example of an observation from the original dataset

```
df.head()
✓ [119] 34ms
```

	Price	District	City	Town	Type	EnergyCertificate	GrossArea	TotalArea	Parking	HasParking	...	Elevator	ElectricCarsCharging	TotalRooms	Number
0	780000.0	Vila Real	Valpaços	Carrazedo de Montenegro e Curros	Farm	NC	200.0	552450.0	0.0	False	...	False	NaN	NaN	
1	223000.0	Faro	São Brás de Alportel	São Brás de Alportel	Apartment	A+	NaN	81.0	1.0	True	...	True	NaN	2.0	
2	228000.0	Faro	São Brás de Alportel	São Brás de Alportel	Apartment	A+	NaN	108.0	1.0	True	...	True	NaN	2.0	
3	250000.0	Faro	São Brás de Alportel	São Brás de Alportel	Apartment	A+	NaN	114.0	1.0	True	...	True	NaN	2.0	
4	250000.0	Faro	São Brás de Alportel	São Brás de Alportel	Apartment	A+	NaN	114.0	1.0	True	...	True	NaN	2.0	

5 rows × 25 columns

```
df.head()
✓ [119] 34ms
```

ing	HasParking	...	Elevator	ElectricCarsCharging	TotalRooms	NumberOfBedrooms	NumberOfWC	ConservationStatus	LivingArea	LotSize	BuiltArea	NumberOfBathrooms
0.0	False	...	False	NaN	NaN	NaN	NaN	NaN	120.0	NaN	NaN	0.0
1.0	True	...	True	NaN	2.0	NaN	NaN	NaN	81.0	NaN	NaN	2.0
1.0	True	...	True	NaN	2.0	NaN	NaN	NaN	108.0	NaN	NaN	2.0
1.0	True	...	True	NaN	2.0	NaN	NaN	NaN	114.0	NaN	NaN	0.0
1.0	True	...	True	NaN	2.0	NaN	NaN	NaN	114.0	NaN	NaN	2.0

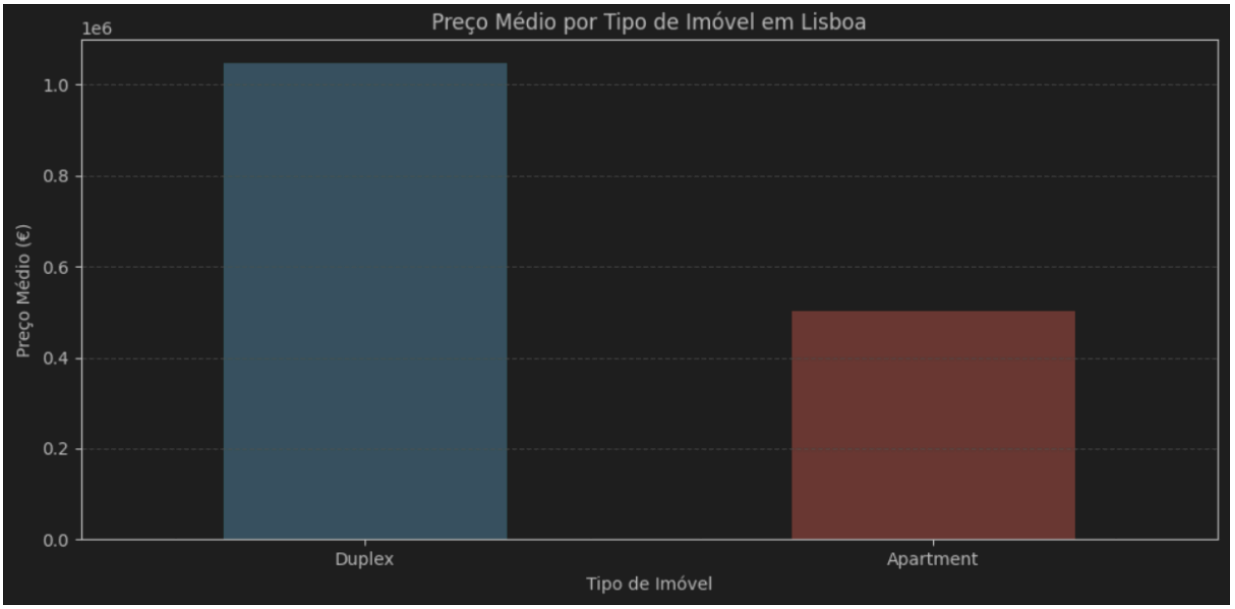
Annex 2 - Example of an observation from the dataset after data processing

	Price	City	Town	Type	TotalArea	HasParking	Elevator	TotalRooms	NumberOfBathrooms	Age
757	200000.0	Sintra	Algueirão-Mem Martins	Apartment	79.0	True	True	2.0	1.0	24.0
1619	595000.0	Lisboa	Misericórdia	Apartment	110.0	False	False	4.0	2.0	87.0
1640	315000.0	Lisboa	Ajuda	Apartment	68.0	False	True	2.0	1.0	61.0
1682	100000.0	Azambuja	Azambuja	Apartment	208.0	False	False	2.0	1.0	46.0
2202	310000.0	Lisboa	Ajuda	Apartment	68.0	False	True	2.0	1.0	61.0

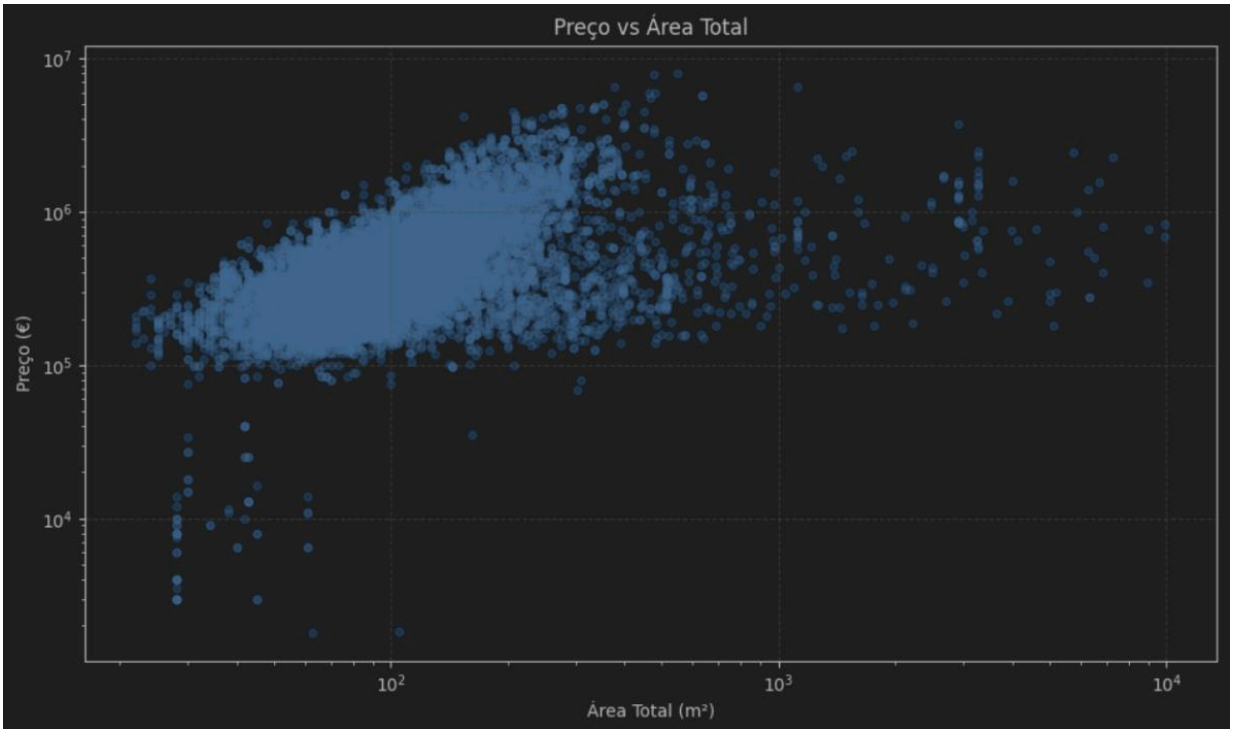
Annex 3 - Performance metrics of the predictive model

```
R²: 0.89
MAE: 70,818.69 €
RMSE: 153,036.23 €
MAPE: 15.13%
```

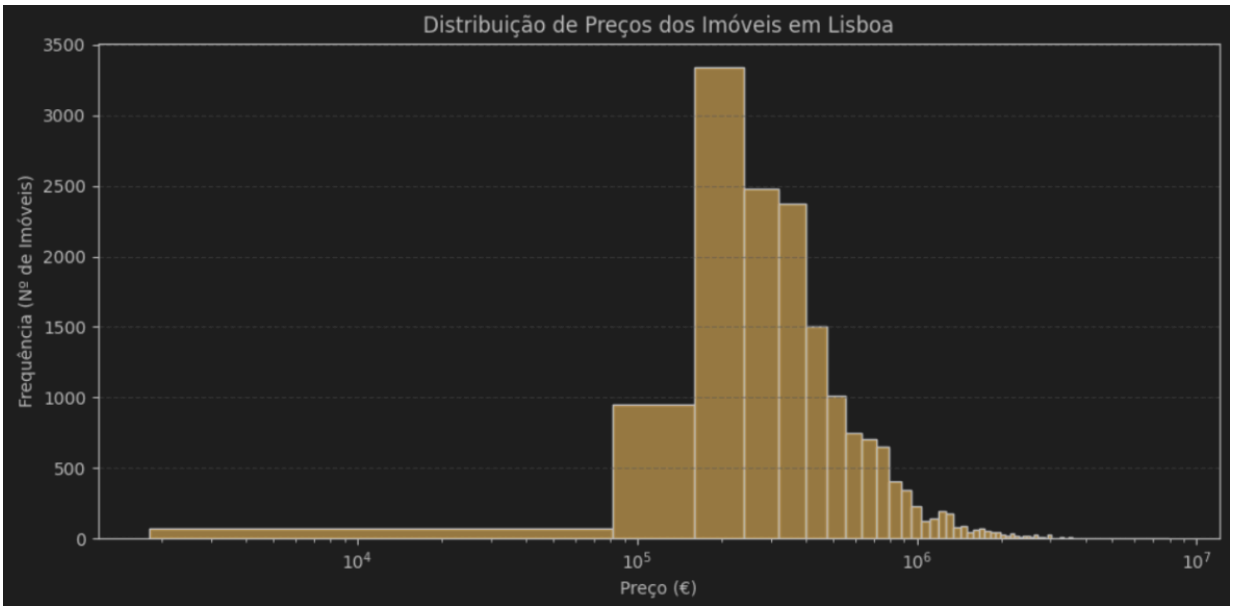
Annex 4 - Average price by urban property type in the district of Lisbon



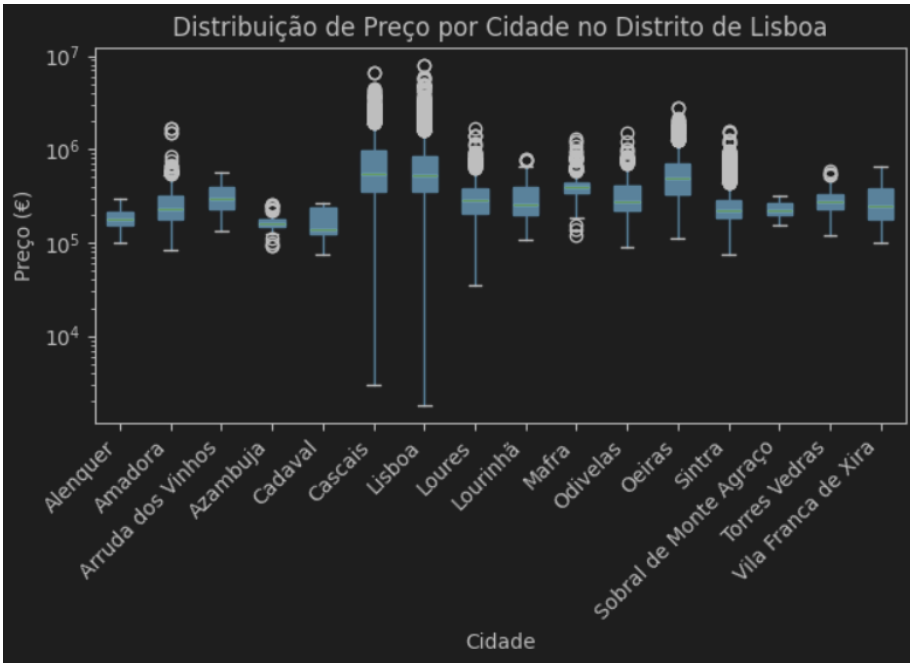
Annex 5 - Relationship between total area and property price (log. scale on the Y axis)



Annex 6 - Distribution of urban property prices in the district of Lisbon (log. scale on the X axis)



Annex 7 - Distribution of prices by city in the district of Lisbon (boxplot with log. scale)



Annex 8 - Correlation matrix between numerical variables

